



**UNIVERSITI TEKNOLOGI MARA
TEST**

COURSE	: IMAGE PROCESSING
COURSE CODE	: CSC 566
TIME	: 2 HOURS
SEMESTER	: MARCH – AUGUST 2026

INSTRUCTIONS TO CANDIDATES:

1. This question paper consists of **FIVE (5)** questions.
2. Answer **ALL** questions. **Write your answers in your own paper. Scan or snap and submit immediately to the provided link given.**

STUDENT DETAILS:

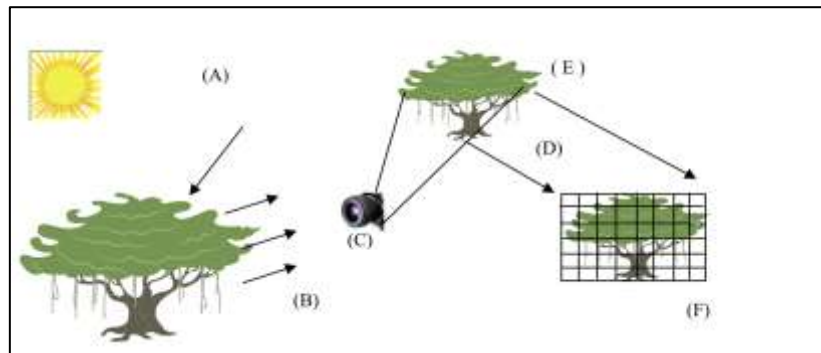
MATRIC NO		CLASS	
FULL NAME			
LECTURER			
MARKS	QUESTION 1		
	QUESTION 2		
QUESTION 3			
QUESTION 4			
QUESTION 5			
	TOTAL	/ 100	

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 8 printed pages

QUESTION 1 (20 MARKS)

- a) Image acquisition is the beginning process in image processing to produce a digital image. **Figure 1** shows the elements that are involved in the digital image production. Label each of the elements.

**Figure 1**

(6 marks)

- b) From the labeling that you have made in (a), derive the equation for a digital image. (3 marks)
- c) State **ONE (1)** image processing application. Discuss the benefit of the selected image processing application to the society. (3 marks)
- d) **Figure 2** shows a checkerboard effect on an image. Discuss **TWO (2)** reasons that contribute to the checkerboard effect on the image.

**Figure 2.** Checkerboard effect on the image

(4 marks)

- e) Differentiate between nearest neighbor interpolation and bilinear interpolation.

(4 marks)

QUESTION 2 (20 MARKS)

a) Fill in the blanks.

(10 marks)

- i. Histogram is used to represent the _____ of grayscale intensities in an image.
- ii. Histogram cannot represent the _____ information in an image.
- iii. In histogram stretching, the _____ and _____ brightness values from the histogram need to be identified. Next, the transformation will uniformly _____ this range to fill the full range.

b) Given a grayscale image as presented in **Figure 3**. Enhance the image using histogram equalization.

(8 marks)

0	15	20	60	35	25
30	30	35	35	70	50
30	50	15	35	70	25
0	0	20	20	70	0
70	45	45	45	35	0
50	0	60	0	50	0

Figure 3. A grayscale image.

c) Draw the equalized image.

(2 marks)

QUESTION 3 (20 MARKS)

a) Briefly define sampling and quantization.

(2 marks)

b) Fill in the missing information in the table below to match the Image Type with its corresponding Data Type.

(4 marks)

Image Type	Data Type
i)	{0, 1}
ii)	{0, 1}, uint8, or uint16
iii)	m-by-3 colour map matrix
iv)	m-by-n-by-3 matrix

c) Write TRUE or FALSE.

(5 marks)

STATEMENTS	TRUE / FALSE
Bit depth determines the gray levels allowed for each pixel.	
Upon reducing the quantization rate to save storage space, false contouring may occur in the resulting image.	
Sampling determines the biggest discernible detail in the image.	
Zooming is undersampling, while shrinking is oversampling for a digital image.	
Zooming produces checkerboard effects.	

d) **Figure 4** shows the Matlab computing platform:

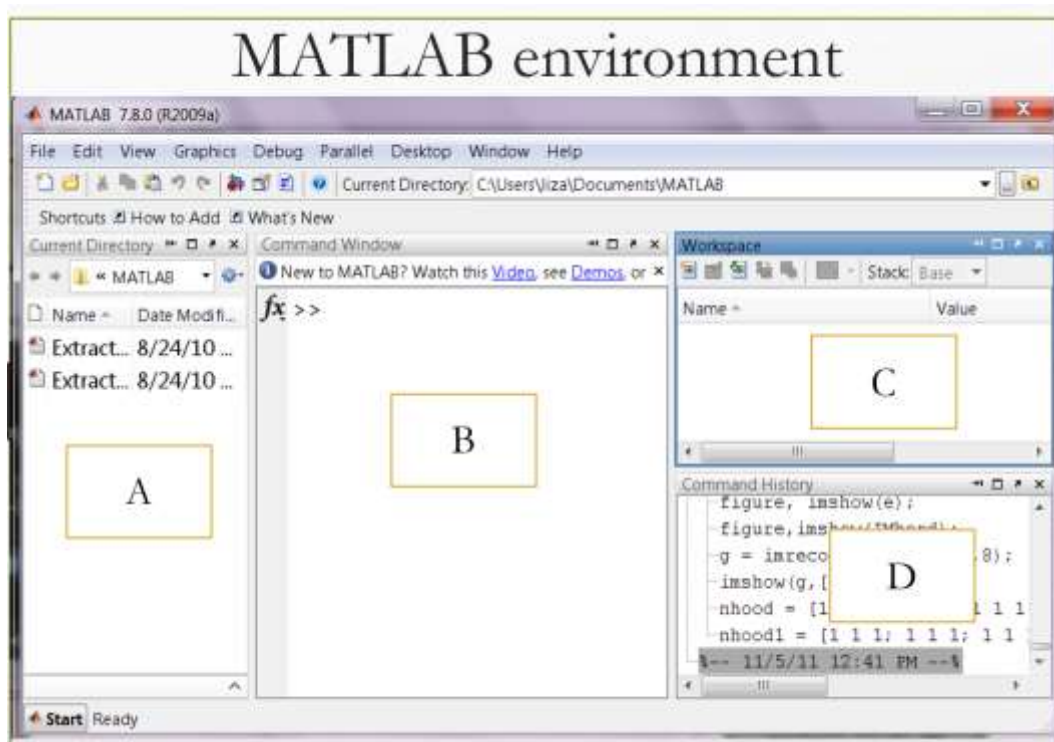


Figure 4. Matlab

List what is the name of the area labelled **A**, **B**, **C**, and **D**?

(4 marks)

- e) Fill the appropriate column with a built-in function that you learn in MATLAB.
(See the example shown)

(5 marks)

Function	Purpose
Example Answer: im2uint16	Image array to 16-bit unsigned integers
	Indexed image to intensity image
	Indexed image to RGB image (MATLAB)
	Intensity image to indexed image
	RGB image or colourmap to grayscale
	RGB image to indexed image
	Matrix to intensity image
	Image to binary image by thresholding
	Image array to double precision
	Image array to 8-bit unsigned integers
	Display multiple images or plots within a single figure window

QUESTION 4 (20 MARKS)

- a) Solve the logical operations below:

(6 marks)



- i. A AND B
- ii. A OR B
- iii. NOT B

- b) Given an image size 6 x 6 as presented in **Figure 5(a)** and a weighted filter for average filtering as illustrated in **Figure 5(b)**.

99	166	12	166	44	105
210	105	210	99	210	65
44	140	105	79	12	99
105	12	99	210	166	210
166	88	210	127	105	166
99	44	166	12	99	44

Figure 5(a)

1/16	1/16	1/16
1/16	1/16	1/16
1/16	1/16	1/16

Figure 5(b)

- Perform average filtering using weighted filter on pixel 140 and pixel 65. (6 marks)
- Perform histogram stretching on pixel 127. Given, $J = 255 * \frac{I - I_{min}}{I_{max} - I_{min}}$ (2 marks)
- Perform image negative on pixel 79. Given, $T = L - 1 - f(x, y)$ (2 marks)
- Perform median filtering on pixel 88. (2 marks)

v. Perform log transformation on pixel 1568.3.

(2 marks)

932.2	812.5
-734.9	556.1
123.4	1568.3
4321.1	-1683.5

QUESTION 5 (20 MARKS)

- a) Explain **ONE (1)** step that involves extracting image components which is useful in the representation and description of shape.

(2 marks)

- b) **Figure 6(a)** shows an original handwriting binary image. Name and describe the morphological operator that may be applied to produce the image shown in **Figure 6(b)**.

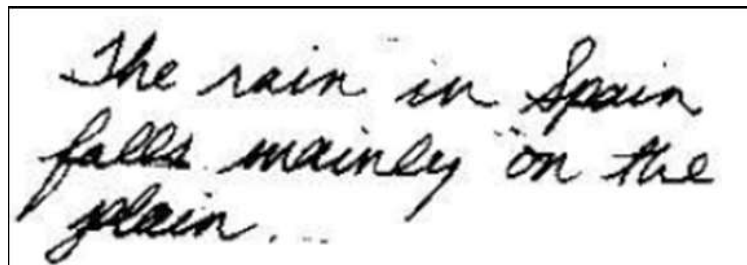


Figure 6(a)

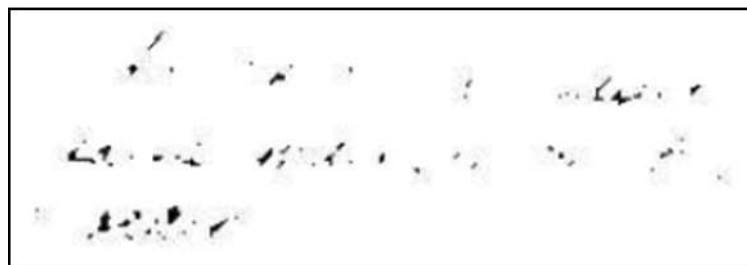


Figure 6(b)

(3 marks)

- c) Sketch the set $(A \cap B) \cup (A \cup B)^c$ based on the given **Figure 7** shown below.

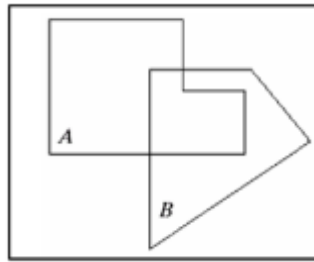


Figure 7

(5 marks)

- d) Perform the following morphological operations using the given input image in **Figure 8(a)** and the structuring element in **Figure 8(b)**. The origin is shown as the circled element.

1	1	1	1
0	0	1	1
0	0	0	1
0	1	1	1

Figure 8(a)

①	1	1
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Figure 8(b)

- i. Erosion

(4 marks)

- ii. Opening

(6 marks)

END OF QUESTION PAPER

GOOD LUCK DEAR STUDENTS!